

SYSTEM GOVERNANCE & ASSURANCE [0.1–1 Hz]

Shared Autonomy (§14) · Verification (§15) · Safety Cases (§16) · Frontier Limits (§17)

[SYSTEM 2: 1-50 Hz · STOCHASTIC DELIBERATION]

1. DATA & INGESTION

Demos & Wear (§05)

MIPI CSI-2 · Serial DMA Ring Buffers

2. SPATIAL PERCEPTION

$o_t \rightarrow z_t$ Tokens (§08)

ViT Encoders · DINOv2 · 3D Affordances

3. TEMPORAL MEMORY

$z_t \rightarrow b_t$ Belief (§09)

SE(3) Transform Trees · World Models

5. TRAJECTORY PLANNER (System 1.5 · MPU/NPU)

$L_t \rightarrow \hat{u}_{t:t+H}$ Chunk (§11)

Diffusion Policies · ACT Action Chunking · c^2 Jerk Continuity

4. INTENT & GOALS (System 2 · MPU)

$b_t \rightarrow L_t$ Lease (§10)

Vision-Language Models · Expiring Intent Leases (t_{expire})

PROPOSAL-PERMISSION PRIVILEGE BOUNDARY (GATED WRITE $\hat{u} \rightarrow u^*$)

1 kHz REFLEX ENFORCER (§12)

Control Barrier Filter ($h(\mathbf{x}) \geq 0$) $\rightarrow$ Certified u^* (PWM Compare Register)

SILICON PLACEMENT & BUS QoS (§13)

Crossbar QoS · SRAM Seqlock · Hardware Timers

THE PHYSICAL PLANT (THE BODY) · Matter, Momentum, Friction & Thermal Limits (§01, §02)

Causal Boundary (10–20 kHz Inverter) · Stopping Envelope ($d_{\text{stop}} \leq d_{\text{clear}}$) · Sim-to-Real Gap (§06, §07)

ENDOGENOUS SENSORY SHIFT (o_{t+1})